

University of California, Davis
Department of Economics

PRELIMINARY EXAMINATION FOR THE Ph.D. DEGREE

MACROECONOMICS

July 3, 2025

Directions: The exam consists of six questions. Questions 1,2 concern ECN 200D (Geromichalos), questions 3,4 concern ECN 200E (Cloyne), and questions 5,6 concern ECN 200F (Caramp). You only need to answer five out of the six questions. If you prefer (and have time), you can answer all six questions, and your grade will be based upon the best five scores. Feel free to impose additional structure on the problems below, but please state your assumptions clearly. You have 5 hours to complete the exam and an additional 20 minutes of reading time.

Question 1 (20 points)

Consider a neoclassical growth model in discrete time. There is a large number of identical households normalized to 1. Each household wants to maximize life-time discounted utility

$$U(\{c_t\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t u(c_t), \quad \beta \in (0, 1),$$

where u is twice continuously differentiable and bounded, with $u'(c) > 0$, $u''(c) < 0$, $u'(0) = \infty$, and $u'(\infty) = 0$. As is standard, households own the capital stock (they make the investment decision), and they rent it to firms. Each household has one unit of productive time in each period, and they rent these “labor services” to firms.

The main difference of this model compared to the baseline model studied in class concerns production. There is a large number of identical firms each of which has the following production function:

$$F(k, l, \bar{k}) = Ak^a l^{1-a} \bar{k}^{\gamma},$$

where k, l are the amounts of capital and labor rented by the firm, and $\bar{k}$ is the aggregate capital stock in the whole wide economy. Assume that $a \in (0, 1)$ and $\gamma \in (0, 1 - a)$. Thus, here there is a *productive externality* from the rest of the economy: a higher aggregate capital stock increases each firm’s productivity. Naturally, a typical (small) firm takes the aggregate capital stock as given when choosing its inputs.

The rest of the model is standard. Output can be consumed (c_t) or invested (i_t), and we will let $\delta \in (0, 1)$ denote the depreciation rate of capital. Thus, **if we set $\gamma = 0$ this would be identical to the growth model studied in class.**

a) Define a recursive competitive equilibrium for this economy. Be clear about which variables households and firms take as given when they solve their optimization problems. Find the competitive equilibrium steady-state aggregate capital stock as a function of the parameters of the model.

b) Write the social planner’s problem for this economy in recursive form, assuming that the social planner internalizes the externality in production.¹ Characterize the steady-state aggregate capital stock implied by the planner’s problem. Discuss whether that capital stock is higher or lower than the one obtained in the competitive equilibrium (in part (a)) and provide economic intuition for your result.

¹ That is, in the social planner’s problem, the production function is

$$F(\bar{k}, l, \bar{k}) = A\bar{k}^{a+\gamma} l^{1-a}.$$

c) Now introduce a government into the competitive equilibrium that you defined in part (a). The government subsidizes investment expenditures at a proportional rate τ and finances these subsidies by imposing a lump-sum tax T on each household. (Assume that the government's budget has to be balanced in every period.) Define a recursive competitive equilibrium for this economy.

d) For what subsidy rate τ is the competitive equilibrium steady-state aggregate capital stock the same as the steady-state aggregate capital stock in the social planner's problem?

Question 2 (20 points)

This question studies the co-existence of money and credit. Time is discrete with an infinite horizon. Each period consists of two subperiods. In the “day”, trade is bilateral and *partially* anonymous as in Kiyotaki and Wright (1993) (call this the KW market). At “night” trade takes place in a Walrasian or centralized market (call this the CM). There are two types of agents, buyers and sellers, and the measure of both is normalized to 1. The instantaneous utility for buyers is $u(q) + X - H$, and for sellers it is $-q + X - H$, where q is the quantity of special good produced by the seller and consumed by the buyer, X is consumption of the CM good (the numeraire), and H is hours worked in the CM. In the CM, all agents can turn one unit of labor into a unit of good. Define the first-best quantity traded in the KW market as $q^* \equiv \{q : u'(q^*) = 1\}$.

What is new here is that there are two types of sellers. Type-N sellers, with measure $1 - \Lambda$, never accept unsecured credit. Hence, in any meeting with this seller type, the buyer must use money and settle the payment on the spot. In contrast, type-C sellers, with measure Λ , accept unsecured credit, in the form of a promise by the buyer to repay the seller in the forthcoming CM with numeraire good. There is no credit limit, and the buyer can *credibly* promise to repay any amount in the forthcoming CM. We assume that all buyers meet a seller in the KW market. (So that with probability $1 - \Lambda$ that seller is of type-N, and with probability Λ the seller is of type-C.)

Goods are non storable. There exists a storable and recognizable object, fiat money, that can serve as a medium of exchange. Money supply is controlled by a monetary authority, and we consider simple policies of the form $M_{t+1} = (1 + \mu)M_t$, $\mu > \beta - 1$. New money is introduced, or withdrawn if $\mu < 0$, via lump-sum transfers to buyers in the CM. Let ϕ denote the unit price of money (in terms of the numeraire). In KW meetings, buyers and sellers (of either type) negotiate over the terms of trade based on Kalai bargaining, with $\theta \in (0, 1)$ capturing the bargaining power of buyers.

a) Describe the CM value function, $W(m, d)$, of a buyer who enters the CM with m units of money and a debt d (a promise to repay a seller of type-C). Similarly, describe the CM value function, $W^S(m, d)$, of a seller who enters that market with m units of money, and who is entitled to a payment of d units of numeraire good (by a buyer).

b) Describe and solve the bargaining problem between a buyer who carries m units of money and a seller of type-N. Do the same for the bargaining problem between a buyer and a seller of type-C.

c) Derive the representative buyer’s objective function (i.e., the function whose maximization will deliver this agent’s money demand).

d) Obtain the first-order condition for maximization of the objective function. Then, focus on a steady-state equilibrium, which here consists of z (real money balances), q (quantity of good traded in meetings with N-type sellers), and q_C (quantity of

good traded in meetings with C-type sellers). Describe the three equilibrium variables as functions of the model's parameters, including Λ and the "Fisher rate" i .

So far we assumed that the measure of sellers who accept credit, Λ , is exogenous. In what follows, we will derive Λ endogenously, as the outcome of the sellers' optimal behavior. We will assume that in the beginning of each period sellers have the option to pay a cost κ that allows them to accept credit.² Notice that under this new setting, the measure of (type-C) sellers who accept credit, Λ , is just the measure of sellers who decide to pay κ , and, similarly, $1 - \Lambda$ is the measure of sellers who decide to not pay κ and remain type-N. Obviously, a seller will choose to pay κ if the (net) benefit from becoming a type-C seller exceeds the benefit of not paying κ (thus remaining a type-N). In what follows, we will study the problem of the representative seller, who takes as given the behavior of other sellers in the economy and responds optimally. To that end, let $\Lambda \in [0, 1]$ be the representative seller's belief about the fraction of fellow sellers who have decided to become type-C.³

e) What is the per period (steady state) net benefit, call it S_C , of the representative seller, if she decides to pay κ and become type-C? (Note: it should be equal to the steady state surplus of a C-type seller minus κ .) Does S_C depend on the belief Λ ?

f) What is the per period (steady state) benefit, call it S_N , of the representative seller, if she does **not** pay κ (thus remaining type-N)? Find the sign of $\partial S_N / \partial \Lambda$ and explain intuitively. Notice that you can answer this without any math: think of a seller who contemplates remaining a type-N. What does this seller win or lose as more fellow sellers become type-C? (Think of what happens to the demand for money.)

g) Suppose the representative seller believes that every other seller has become a type-C (i.e., $\Lambda = 1$). What is her best response, $\lambda^*(1)$?

h) On the other extreme, suppose the representative seller believes that no other seller has become a type-C (i.e., $\Lambda = 0$). What is her best response, $\lambda^*(0)$? To make the math easier, feel free to calculate $\lambda^*(0)$ at the limit as $i \rightarrow 0$.

i) Build on your findings in parts (e)-(h) and discuss the set of *all* possible equilibrium values of Λ . (See footnote 3.) Start the analysis assuming $i \rightarrow 0$, as in part (h), but also make a conjecture about what happens as i increases.

² One interpretation is that κ is the cost of a technology that allows sellers to read credit cards. But, again, the sellers must pay that cost in every period, an assumption that we impose for simplicity.

³ Call the representative seller's best-response function $\lambda^*(\Lambda)$. As in Kiyotaki and Wright (1993), an equilibrium will be a point where the representative seller is behaving optimally and is *indeed* representative, i.e., an equilibrium is a point where $\lambda^*(\Lambda)$ intersects with the 45 degree line.

Question 3 (20 points)

Consider the planner's problem for a real business cycle model with inelastic labor supply and no trend growth. Time is infinite, and the planner seeks to maximize lifetime expected utility from consumption (C):

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} \right) \quad (1)$$

where $0 < \beta < 1$ and $\sigma > 0$.

In this model there are two vintages of capital, “new” K_1 and “old” K_2 , that are used to produce output, Y , according to a constant elasticity of substitution production technology:

$$Y_t = A_t \left(\alpha_1 K_{1,t}^{1-\sigma} + \alpha_2 K_{2,t}^{1-\sigma} + (1 - \alpha_1 - \alpha_2) N^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (2)$$

where $\alpha_1 > 0$, $\alpha_2 > 0$ and $\alpha_1 + \alpha_2 \leq 1$. Labor, N , is inelastically supplied, so we can assume $N = 1$.⁴

Capital lasts for two periods. New capital available at the start of period $t + 1$ (denoted $K_{1,t+1}$) is created by investment, I_t :

$$K_{1,t+1} = I_t$$

New capital depreciates at rate δ and becomes “old”, where $0 < \delta \leq 1$. In any period t , denote the old vintage of capital at the start of time t as $K_{2,t}$. The relationship between the two vintages is:

$$K_{2,t+1} = (1 - \delta) K_{1,t} \quad (3)$$

A higher value of δ implies a smaller fraction of the new capital stock today will be available tomorrow once it becomes “old” capital.

Total Factor Productivity, A_t , is stochastic. Assume this follows a stationary Markov process with no persistence. In other words, A_t is independent and identically distributed (i.i.d.).

The aggregate resource constraint is $Y_t = C_t + I_t$.

⁴Note, the standard model studied in class with full depreciation ($\delta = 1$) and log utility has $\sigma \rightarrow 1$ (for simplicity, the utility function above omits the -1 from the numerator) and $\alpha_2 = 0$.

- a) Write down the planner's problem in recursive form.
- b) Derive the first order conditions for the planner's problem. Provide some intuition for the Euler equation you derive. (**Hint:** you will need two envelope conditions for the two types of capital).
- c) Solve this model using guess and verify. In particular, find the solved policy functions for output, consumption and investment.
- (**Hints:** Guess that investment is a constant share of output and follow the usual approach. In addition, note that the marginal product for each type of capital $j \in (1, 2)$ takes the form: $\frac{\partial Y}{\partial K_j} = \alpha_j A^{1-\sigma} \left(\frac{Y}{K_j} \right)^\sigma$. Also assume $E(A_{t+1}^{1-\sigma} | A_t) = \mu$ where $\mu > 0$ is a fixed constant.)
- d) Discuss how, and why, a positive TFP shock affects consumption, investment and output in this model. How does the value of δ affect the predictions of the model?
- e) Suppose, you solve a version of this model computationally using a linearization-based technique. Explain one advantage and one disadvantage of using this approach.

Question 4 (20 points)

The U.S. government is currently considering large reductions in government spending, which could potentially lead to a lower public capital stock (for example, through reduced infrastructure spending). Using the New Keynesian model, this question asks you to analyze the macroeconomic impact, and how monetary policy might respond.

As usual, there are a continuum of identical households. The representative household makes consumption and labor supply decisions to maximize lifetime expected utility. In linearized form, the household first order conditions are as follows:

$$E_t \hat{c}_{t+1} - \hat{c}_t = \hat{i}_t - E_t \hat{\pi}_{t+1} \quad (4)$$

$$\hat{w}_t = \hat{c}_t + \psi \hat{n}_t \quad (5)$$

In percentage deviations from steady state: $\hat{c}_t$ is consumption, $\hat{w}_t$ is the real wage, and $\hat{n}_t$ is hours worked. In deviations from steady state: $\hat{i}_t$ is the nominal interest rate and $\hat{\pi}_t$ is inflation ($\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$; $\hat{p}_t$ is the percentage deviation of the price level from steady state). The coefficient $\psi = \frac{N}{1-N}$. N denotes steady state hours worked, $0 < N < 1$ and $\psi > 0$.⁵

The production side of the model is similar to the standard New Keynesian environment, except for the inclusion of public capital in the intermediate goods production process. Monopolistically competitive intermediate goods firms produce an intermediate good using labor as usual and public capital which is as an “external input” and not chosen by firms. The production function for product variety j is given by:

$$y_t(j) = K_{G,t}^\chi n_t(j) \quad (6)$$

Public capital $K_{G,t}$ is common across all firms and evolves as follows:

$$K_{G,t} = (1 - \delta)K_{G,t-1} + G_t \quad (7)$$

G_t is government investment. For simplicity, assume full depreciation $\delta = 1$ and $K_{G,t} = G_t$. χ is the elasticity of output with respect to public capital and $0 \leq \chi \leq 1$.

The rest of the model is standard and intermediate goods firms face a probability that they cannot adjust their price each period (the Calvo pricing mechanism). As usual, intermediate goods are aggregated into a final (homogenous) consumption good by final goods firms using the constant elasticity of substitution (CES) aggregator.

⁵Household preferences are $\eta \ln C_t + (1 - \eta) \ln(1 - N_t)$ where $0 < \eta < 1$.

The production side of the economy, when aggregated and linearized, can be described by the following set of linearized equilibrium conditions (the production function, the optimal hiring condition for labor and the dynamic evolution of prices):

$$\hat{y}_t = \chi \hat{k}_{G,t} + \hat{n}_t \quad (8)$$

$$\hat{w}_t = \hat{m}c_t + \chi \hat{k}_{G,t} \quad (9)$$

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda \hat{m}c_t \quad (10)$$

The resource constraint is:
$$\hat{y}_t = \gamma_c \hat{c}_t + \hat{g}_t \quad (11)$$

Monetary policy follows a simple Taylor Rule:
$$\hat{i}_t = \phi_\pi \hat{\pi}_t \quad (12)$$

Government investment (linearized) follows:
$$\hat{g}_t = \rho \hat{g}_{t-1} + e_t \quad (13)$$

e_t is i.i.d. $\hat{g}_t$ is the change in government investment relative to steady state as a proportion of steady state output: $\hat{g}_t = \frac{G_t - G}{Y}$. This implies that, with 100% depreciation each period, $\hat{k}_{G,t} = \frac{1}{\gamma_g} \hat{g}_t$ where γ_g is the steady state share of government investment in output and assume $0 < \gamma_g < 1$. Government investment is financed entirely by lump sum taxes on households. In percentage deviations from steady state: $\hat{m}c_t$ is real marginal cost, $\hat{y}_t$ is output and $\hat{k}_{G,t}$ is public capital. γ_c is the steady state share of consumption in output. λ is a function of model parameters, including the degree of price stickiness.⁶ Assume that $\phi_\pi > 1$, $0 < \rho < 1$.

a) Using the relevant equations above show that, with flexible prices, the natural level of output can be written as

$$\hat{y}_t^n = \Gamma \hat{g}_t \quad (14)$$

where

$$\Gamma \equiv \frac{1}{1 + \psi \gamma_c} \left(1 + \frac{\gamma_c (1 + \psi) \chi}{\gamma_g} \right)$$

Why does the response of the natural level of output depend on the value of χ ?

b) The full sticky price model can be simplified to 3 equations:

$$E_t \tilde{y}_{t+1} - \tilde{y}_t = \gamma_c (\hat{i}_t - E_t \hat{\pi}_{t+1} - \frac{1}{\gamma_c} (1 - \Gamma) (1 - \rho) \hat{g}_t) \quad (15)$$

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \kappa \tilde{y}_t \quad (16)$$

$$\hat{i}_t = \phi_\pi \hat{\pi}_t \quad (17)$$

⁶ $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}$ where θ is the probability that a firm cannot adjust its price.

plus the stochastic process for government investment (equation (13)). $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$ is the output gap and $\kappa = \frac{1+\psi\gamma_c}{\gamma_c}\lambda$.

Using the method of undetermined coefficients, find the response of the output gap and inflation to an exogenous decrease in government investment when prices are sticky and monetary policy follows the Taylor Rule above. To do this, guess that the solution for each variable is a linear function of the government investment shock $\hat{g}_t$.

c) Discuss how, and why, a fall in public investment affects the output gap and inflation in this model. Take care to explain how your answer depends on the value of χ (**Hint:** you may want to consider two cases: $\chi = 0$ vs. $\chi = 1$, and also consider the expression for Γ).

d) According to the monetary policy rule in equation (12), how will the monetary policymaker adjust the nominal interest rate following a fall in public investment? Does this result depend on the value of χ ? Provide some intuition for your result.

e) Now let's consider whether monetary policy could do better than in part (d). Can you think of a policy rule that would implement full stabilization of inflation and the output gap in this model? Explain. (**Hint:** you do not need to derive anything. You should be able to answer this from your knowledge of the model.)

Question 5 (20 points)

Consider an economy populated by a representative, infinitely-lived consumer. The consumer owns a technology with which she produces output (Y) using capital (K) and land (L) according to the production function:

$$Y = F(K, L),$$

where $F(\cdot)$ is increasing, concave and differentiable. Capital fully depreciates after its use ($\delta = 1$).

The consumer is endowed with K_0 units of capital and $L_0 = 1$ units of land at $t = 0$, and has access to an internal land market; i.e., she can buy and sell land in the local market at a price Q_t . The consumer also has access to an international financial market: the consumer can borrow B_t units of consumption goods from international markets at time t , at an interest rate $r > 0$.⁷ However, they are constrained on how much they can borrow:

$$B_t \leq \bar{B} \quad \text{for all } t, \quad (18)$$

where $\bar{B}$ is an exogenously given borrowing limit. The consumer is born with a debt of $B_0 < \bar{B}$.

The consumer has preferences over consumption streams according to the utility function

$$U = \sum_{t=0}^{\infty} \left(\frac{1}{1+\rho} \right)^t \ln(C_t) \quad (19)$$

where we assume that $\rho = r$. The consumer chooses sequences of consumption $\{C_t\}_{t=0}^{\infty}$, capital $\{K_{t+1}\}_{t=0}^{\infty}$ and land purchases $\{L_{t+1}\}_{t=0}^{\infty}$ to maximize (19) given prices $\{Q_t\}_{t=0}^{\infty}$ subject to the budget constraint:

$$C_t + K_{t+1} + Q_t L_{t+1} + B_t(1+r) \leq F(K_t, L_t) + B_{t+1} + Q_t L_t, \quad (20)$$

and the borrowing constraint (18). So, the consumer must finance consumption C_t , investment K_{t+1} , interest payments on debt $B_t(1+r)$ and land purchases $Q_t L_{t+1}$ using output, borrowing more funds (B_{t+1}) and selling the land they own at the price Q_t . To close the economy, assume that the total supply of land remains constant over time and equal to one; i.e., $L_t = 1$ for all $t = 1, 2, \dots$

a) State the consumer's problem. Characterize the solution with first-order conditions. Let λ_t be the Lagrange multiplier associated to the budget constraint (20),

⁷The international financial markets allow the consumer to save or borrow as much as she wants at the given interest rate r . Technically, this means that the market for bonds *does not need to clear* in a competitive equilibrium. In contrast, we assume that the market for land is local, i.e., it needs to clear in equilibrium.

and μ_t the Lagrange multiplier associated to the borrowing constraint (18). Define a competitive equilibrium for this economy. **Hint:** Remember that the financial markets are international, so you do not need to include a market clearing condition for bonds.

b) A steady-state equilibrium is an equilibrium in which $C_t = C_{ss}$, $B_t = B_{ss}$, $Y_t = Y_{ss}$, $Q_t = Q_{ss}$ and $K_t = K_{ss}$ for all t . Using the definition of a competitive equilibrium and the FOC's of the consumer's optimization problem, characterize a steady-state equilibrium, given a borrowing level $B_{ss} < \bar{B}$. Do K_{ss} , Y_{ss} and Q_{ss} depend on the level of B_{ss} ?

c) Assume that $K_0 = K_{ss}$, $B_0 = B_{ss} < \bar{B}$, and $L_0 = 1$ (so the economy is at its steady state from period $t = 0$). Suppose that there is an unexpected positive shock on the debt limit; i.e., $\bar{B}' = \bar{B} + \Delta$, with $\Delta > 0$. Does this shock have any effect on consumption, output and land prices? What if $\bar{B}' = \bar{B} - \Delta$ with $\Delta \in (0, \bar{B} - B_0)$? Would it change your results if there was no borrowing constraint?

d) Suppose that $\bar{B}' = \bar{B} - \Delta$, and that $\Delta > \bar{B} - B_0$ (so that the initial borrowing level exceeds the borrowing limit $\bar{B}$). What happens with equilibrium output, Y_t , and the equilibrium price of land, Q_t ? **Hint:** Guess that there exists an equilibrium in which the borrowing constraint (18) binds every period, i.e. $\mu_t > 0 \forall t$. First, show that in such an equilibrium, $\{C_t\}_{t=0}^{\infty}$ increases over time. Then, show that $K_{t+1} < K_{ss}$ for all t . Finally, iterate forward the optimality condition for land to obtain the price of land as a function of the future path of consumption and capital.

e) Suppose now that instead of the borrowing constraint (1), consumers must now collateralize their loans with the value of their land holdings. Namely,

$$B_{t+1} \leq Q_t L_{t+1}.$$

Explain in words how this economy differs from the one in part d). In particular, explain whether the endogenous response of the price of land, Q_t , amplifies or dampens the initial shock. Is this an example of fire sales?

Question 6 (20 points)

This problem analyzes the effect of future financial constraints on current investment decisions.

Consider an economy with two dates, denoted by $t = 1, 2$. There are two goods: consumption and capital. There is a continuum of entrepreneurs and a continuum of consumers. All individuals have linear utility and consume only in period 2, so $U = E[c_2]$. There is a fixed supply of capital $\bar{k}$, initially owned by the consumers. Consumers are endowed with a large amount of consumption good in each period and they can store this good between dates, such that if they store one unit of the consumption good in $t = 1$ they get one unit of the good in $t = 2$. The entrepreneurs have access to a linear technology that produces A units of consumption good in period 2 per unit of capital they own. The consumers have access to a concave technology $G(\tilde{k})$, where $\tilde{k}$ denotes the capital owned by consumers in period 2. Assume $\lim_{k \rightarrow 0} G'(k) = \infty$ and $G'(\bar{k}) = 0$. The entrepreneurs enter period 1 with a given net worth N_1 in terms of consumption goods, and they do not have access to the storage technology. Assume agents can trade a risk-free bond b_2 that pays an interest rate r .

a) Argue that if the consumers' endowment is sufficiently large, the gross rate of return on the risk-free bond is equal to 1 (i.e., the net return, r , is zero).

b) Suppose that entrepreneurs face no borrowing constraints. State the optimization problems of an entrepreneur and a consumer. Show that the equilibrium capital price is $q_1 = A$ and the entrepreneurs buy k^* , where $G'(\bar{k} - k^*) = A$.

From now on, assume that $N_1 < Ak^*$.

c) Suppose that the entrepreneurs cannot borrow at all, so $q_1 k_2 \leq N_1$. Find the equilibrium price and allocation, show that $q_1 < A$ in equilibrium and that the expected utility of the entrepreneur is $\frac{A}{q_1} N_1$. Show that q_1 is increasing in N_1 .

Let's add one period prior to period 1, period 0. Now the economy has three dates $t = 0, 1, 2$. In period 0 the entrepreneurs have initial net worth N_0 (in consumption goods). Then they borrow b_1 from the consumers and buy capital k_1 at the price q_0 (capital never depreciates). The technology in period 1 is the same as in period 2: consumers produce using the concave technology $G(k)$ and entrepreneurs have the linear technology. However the productivity of the entrepreneurs at date 1 is a random variable a distributed on $[\underline{a}, \bar{a}]$ with $E[a] = A$. The productivity in period 2 is fixed at A .

The entrepreneurs can only issue non-state contingent, risk-free bonds, so b_1 has to satisfy

$$\underbrace{(q_1(a) + a)k_1 - b_1}_{\equiv N_1(a)} \geq 0 \quad \forall a. \quad (21)$$

Note that now the asset price q_1 is in general a function of the shock a and we denote it by $q_1(a)$.

d) Assume that the entrepreneurs' initial net worth N_0 is sufficiently large so that (21) is not binding, and that they face no borrowing limit in period 1. Show that asset prices are $q_0 = 2A$ and $q_1(a) = A$ for all a , and that the capital owned by the entrepreneurs is constant at k^* .

e) Now suppose that the entrepreneurs cannot borrow at date 1 and that $N_1(a) < Ak^*$ for all a , but they are unconstrained at date 0 (i.e., constraint (21) is not binding). Argue informally that now the equilibrium asset price q_1 is weakly increasing in a . **Hint:** Check your answer to question c).

f) Keep the assumption from question e). Write the entrepreneur's problem at date 0. Derive the first-order conditions for the entrepreneur's problem. Derive an expression for the equilibrium price q_0 as a function of $q_1(a)$. **Hint:** Remember that now $N_1(a) = (q_1(a) + a)k_1 - b_1$.

g) Prove that

$$q_0 \leq E[a + q_1(a)] \quad (22)$$

with strict inequality if $q_1(a)$ is not constant. Why is the date 0 price of capital below its present discounted value, despite the fact that entrepreneurs have linear utility and are unconstrained? Why do entrepreneurs not borrow and leverage their investment more to profit from the positive expected return? **Hint:** Remember that if x and y are two random variables, then $cov(x, y) = E[xy] - E[x]E[y]$.